Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

III SEMESTER B. TECH (ELECTRICAL & ELECTRONICS ENGINEERING)

END SEMESTER EXAMINATIONS, NOVEMBER-DECEMBER 2024

DIGITAL SYSTEM DESIGN [ELE 2122]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 01December 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.
- Steps must be clearly shown

1A Determine the minimal sum of products Boolean expression for the given function using QM/Tabulation Technique. 3

$$F(A,B,C,D) = \sum m(2,3,4,5,13,15) + d(8,9,10,11)$$

1B A museum has 4 rooms, each with a motion sensor m0, m1, m2, and m3 respectively, that outputs 1 when motion is detected. After the museum has been closed for the public, the only person in the museum is one security guard walks from room to room. Realize the digital circuit to obtain the following outputs, using active low 4:16 decoder and gates with the following specifications: 3

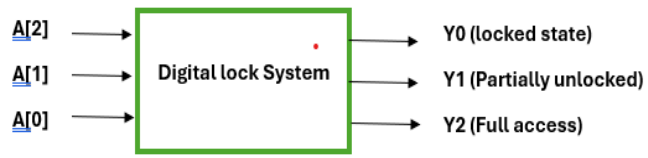
i. Circuit should sound an alarm (output is 1) if motion is ever detected in more than one room at a time (meaning there must be an intruder or intruders in the museum).
ii. Circuit should detect whether the guard is properly patrolling the museum.

1C Obtain a minimal SOP form for the given function, 4

f(a, b,c,d,e)= ∑m(1,4,6,7,9,15,16,18,19,22,24,25,28,29)+ d(2,3,8,17,26,27,31)

using VEM. Consider e as MEV

2A Design a 2-to-4 decoder using gate-level modeling in Verilog HDL. Construct a 3-to-8 decoder by combining two 2-to-4 decoders component to design as a digital lock system as shown in the following block diagram. 4

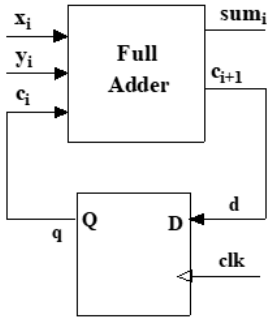


The digital lock system works as follows:

A[2]	A[1]	A[0]	Output
0	0	0	Y0
0	0	1	Y1
0	1	0	Y2

2B Design a 4-to-1 multiplexer (MUX) based digital circuit to manage data selection from four weather sensors: temperature (I0), humidity (I1), wind speed (I2), and rainfall (I3). Consider sensor data width as 8 bits. Use a 2-bit selector (S1, S0) to control which sensor's data is routed to the output. Write Verilog HDL code to implement the 4-to-1 MUX, capable of selecting data from the four sensor inputs. 3

2C Develop the behavioral Verilog code for Full adder and D Flip Flop. Use them as components and write the structural Verilog HDL code for the circuit given below.



3

3A Develop presettable asynchronous up counter which counts the sequence 9-10-11-12-9-…..using JK flip flops. 3

3B Develop a 4-bit Johnson counter using

a) IC 74194 (universal shift register IC) 4

b) Verilog HDL code with behavioral modeling style

3C Convert JK flip flop to AB flipflop. Truth Table of AB Flip-flop is given below.

Truth Table for AB Flip-Flop

A	B	Q+ (next state)
0	0	Q (remains unchanged)
0	1	0 (sets to 0)
1	0	Q' (toggles)
1	1	1 (sets to 1)

3

4A Design synchronous counter which generates the pattern 7 – 5 - 3- 1-7-….using D flip flops. 3

4B A digital system has one input x and one output Z. The output Z=1 occurs every time the input sequence 001 or 011 is detected. Develop a Verilog HDL code using behavioral modeling to detect the pattern using mealy state machine. 4

4C Evaluate the following for Verilog HDL operators, if A = 4'b1100 and B= 2'b10

1. {3{A}, B [1], 2'b10}
2. ^A
3. A || B

3

5A Develop the state diagram for the Microwave oven control with the given specifications:

The oven has a Run push button to start and a Timer that determines the cooking period. Cooking can be interrupted at any time by opening the oven Door. After closing the Door cooking continues. Power should be on to be in cooking state. Cooking is terminated when the Timer elapses. When the Door is opened a Lamp inside the oven is switched on; when the Door is closed the Lamp goes off.

The control system has the following inputs:

- Run push button - when activated starts cooking,
- Timer - while this runs the cooking is enabled,
- Door sensor - can be true (door closed) or false (door open).

3

And the following outputs:

- Power - can be true (power on) or false (power off),
- Lamp - can be true (lamp on) or false (lamp off).

The behavior of the microwave oven control is determined by the Run push button, Timer and Door sensor.

{Assume appropriate state machine and input – output conditions, Mention it in the solution}

5B Describe the IC fabrication steps in detail. 3

5C Construct 2 input NOR gate using CMOS logic families. Hence develop the Verilog HDL code for 2 input NOR gate using switch level logic. 4